Week 8 – Department-Specific Discussions (Lean Notes)

Lectures covered: L32 to L48 (17 talks): 16 IIT Madras departments interviewed by Prof. Prathap Haridoss, then a panel of research scholars (with Prof. S. Sankaran). ⭐ = tested in the Week 8 assignment. Every talk answers the same questions (old areas, new areas, settling in, jobs, success, how often to meet the guide, advice), so each card follows that order. Most of the assignment (Q1–3, Q5–10) tests general research-methods or general-knowledge facts that no talk teaches; card 18 collects them and flags that.

1. Concept cards (in the order taught)

1. Applied Mechanics (L32, Prof. Mahesh Panchagnula)

- **Areas:** fluid mechanics, solid mechanics, biomechanics and biomedical. Faculty come from 8 different UG backgrounds. New: fluid-structure interaction, **cell mechanics**, joining electrical and mechanical; enabled by fast computers and tools (electron microscopy, nano-indentation, atomic force microscopy). Old tools still solve **India-scale problems** (DRDO, ISRO).
- **Starting out:** in class you get 3 of 4 variables; in research you get 1, must **justify your own assumptions** and **define your own area**.
- **Jobs and success:** GE, Boeing, Mercedes, but **academia is India's biggest employer**. Success = growth in confidence and skill, rewarded by **peers outside IIT**.
- **Guide and advice:** frequent at first, then a rhythm, weekly group meetings; conflicts have a professional mechanism. Open mind; research is "a walk through uncharted territory".

2. Chemical Engineering (L33, Profs Abhijit Deshpande and T. Renganathan)

- **Areas:** fluid flow, heat and mass transfer, residence time distribution, multiphase systems, reaction engineering, control; old topics on new problems (fuel-cell flooding, microfluidics, microwave heating) with finer tools (PIV, CFD, lattice Boltzmann). **Process intensification** = reaction and separation in one smaller unit. New: conducting polymers, nano-electrodes, microbes making nanoparticles, semiconductor and food processing.
- **Industry:** 13–15 of the 25 "Make in India" sectors; molecules stay at the centre.
- **Starting out:** more rigour, MATLAB and CFD; labs need **precision, accuracy, repeatability**. Do not start from a UG course or chase buzzwords like "nano".
- **Success and jobs:** explain work at different levels, know its limits, work independently, publish; sign of success: **you know more than your guide**. Jobs: post-doc, academia, R&D, start-ups.
- **Guide and advice:** weekly at first, then fewer, and meet when **results are missing for long**. Be strong in fundamentals.

3. Civil Engineering (L34, Prof. Ravindra Gettu)

- **Areas:** structural, geotechnical, environmental, water resources, transportation, construction management, concrete, **building physics**. New: polymers in construction, IT modelling, the **boom in environmental engineering**.
- **Starting out:** students trained for known answers cannot analyse new situations; about the first year goes to fixing this.
- **Jobs and success:** about **90% of civil PhDs teach**; MS and M.Tech go to industry; some to IAS, IES, IFS. Success = analysing results and deciding.

- **Guide and advice:** about weekly (daily shows a lack of independence). Ask "how will my thesis make the world better?"

4. Computer Science and Engineering (L35, Prof. V. Kamakoti)

- **Areas:** hardware, software (system software, applications, tools), theory. New: **Internet of Things**, data science, machine learning, **information security** (a hacked pacemaker is "cyber murder"), **multi-core** (clock capped near 3–3.5 GHz because power grows with frequency). Industry: low-power hardware, **approximate and stochastic computing**, verification.
- **Success:** everything is **cross-layer** (circuit to application): show synergy across layers and that the idea is **feasible and affordable**.
- **Starting out:** learn the formal "language of CS" through Discrete Mathematics and Knuth's Concrete Mathematics. A theory PhD does not restrict jobs.
- **Guide and advice: peer group matters more than guide**; guide at least fortnightly plus short reports. Hard work, teamwork, ethics; the field changes fast.

5. Engineering Design (L36, Prof. Srikanth Vedantam)

- Other departments teach **analysis**; this one teaches the **design process** (materials and controls for automotive and biomedical), cutting across disciplines.
- **Starting out:** first period = courses across many disciplines.
- **Success:** turn a **vague problem into a specific one**, state assumptions, choose tools, **defend** conclusions. Industry makes incremental tweaks; academia applies fundamental science.
- **Jobs and guide:** mostly industry R&D, academia rising. Frequent meetings early (making the problem specific is hardest), later even monthly; **no spoon-feeding**. Be interdisciplinary; keep the big picture.

6. Humanities and Social Sciences (L37, Prof. Rajesh Kumar)

- **Areas:** economics, literature, linguistics, philosophy, history, sociology; new: **developmental studies**, China studies, European studies, Indo-German Centre for Sustainability. **No boundaries** (health, policy, technology, ethics, religion); an IIT gives easy access to technology data.
- **Starting out:** few faculty per sub-area, so seek specialists outside; a "virtual department". Linguistics has four cores: **syntax** (sentences), **morphology** (words), **phonology** (sounds; transcript says "phrenology", a slip), **sociolinguistics** (language and society).
- **Industry and jobs:** industry needs good community relations in a multilingual country. MA students mostly go for a **PhD** (5-year integrated course after 12th), civil services, NGOs, marketing.
- **Guide and advice: one meaningful hour a week**; arrive with **75%** of the plan done. Be disciplined, treat it as full-time, enjoy it, read beyond your discipline.

7. Mechanical Engineering (L38, Prof. Babu Viswanath)

- **Areas:** three verticals: **thermal**, **design**, **manufacturing** (end-to-end design). New: alternative fuels and emissions, lasers and high-speed cameras inside engines, embedded systems, mechatronics, nanotechnology.
- **Starting out:** rigid academic process; ~1800 students ("a mini college"); many students unclear about their area, handled case by case with bridge courses. **Project work** has a defined problem; **research** does not.
- **Guide:** he meets scholars **daily**. Research is **long disappointment with brief elation**; effort every day matters; feel free to report lack of progress.
- **Jobs and advice:** GM rule: **B.Tech is told "do this", Master's is asked "would you like to do this?", PhD is asked "what should we do?"**. R&D or teaching. Take an IIT MS or PhD seat.

8. Metallurgical and Materials Engineering (L39, Prof. B. S. Murty)

- **Areas:** started 1959 as Metallurgy: **casting, forming, welding**. Modern twists: friction stir (solid-state) welding, hydroforming, severe plastic deformation, metal foams, neutron diffraction for **hot tearing** (transcript: "heart tearing"). Renamed **Metallurgical and Materials Engineering in 2003**: ceramics,

polymers, composites, electronic, bio and nano materials, high entropy alloys; half the faculty are non-metallic.

- **Starting out:** research = plan, execute, analyse, conclude **independently**; reproducible materials; costly microscopes (**1 to 17 crore**, 1 Å); a semester-long practical TEM course; **ICME** (Integrated Computational Materials Engineering).
- **Jobs and success:** mostly academia; industry R&D rising under Make in India. Success = **impact** on industry or on future researchers.
- **Guide and advice:** weekly at first, then at least monthly (he meets students fortnightly). Enjoy it: discuss research over coffee, go to bed thinking about it, excite a 12th-standard student.

9. Ocean Engineering (L40, Prof. V. Anantha Subramanian)

- **Areas:** emerged after the **1970s oil crisis**; ships, offshore and coastal structures, petroleum. Research with FEA and CFD; new: offshore fatigue, controls, **wave energy** (oscillating water column, backward bent duct buoy, ocean thermal energy conversion). Facilities: towing tank, wave basin, flumes. Industry wants **fast, tested solutions**.
- **Starting out:** students from many branches, small faculty; the **rat race for numbers** is counterproductive; do not straitjacket a student; first semester = courses and peer learning.
- **Guide:** against forced meetings; about weekly is fine; learn from a 30–40 member group.
- **Jobs and advice:** shipyards, design firms, offshore, CFD software. Be versatile, share knowledge, be happy.

10. Management Studies (L41, Dr. Usha Mohan)

- **Six functional areas:** organizational behaviour and HR, finance, marketing, operations and production, information systems, integrative. New: **game theory**, digital marketing, sustainability, analytics.
- **Course structure: core** (research methodology, data analysis, academic writing), **breadth** (3 areas), **depth**; first year = courses.
- **Jobs and success:** analyst roles or academia (new IIMs). Success = **ownership**, an “aha” moment, **choosing journals and conferences wisely**.
- **Guide and advice:** fortnightly to monthly at first, fortnightly at literature review, weekly later, much more at thesis writing. **Passion and tenacity** (3–6 years); translate a business need into an abstract problem, then back.

11. Aerospace Engineering (L42, Prof. S. R. Chakravarthy)

- **Areas: aerodynamics** (with flight mechanics), **structures**, **propulsion**. New: unmanned and micro air vehicles, flapping-wing aeroelasticity, micro thrusters, deep space. Spin-offs: the PC (from the space shuttle), head-up displays, gas and wind turbines.
- **Jobs:** a **phenomenon** person suits academia, an **application** person core jobs, a **skills** person (experimental, computational, analytical) anywhere.
- **Starting out:** after a fixed timetable, free time (fewer courses) must be managed: **168 hours a week**; keep expectations high.
- **Guide and success:** about weekly; peers all the time (“campaign mode” labs, more authors per paper). Success is **personal** (h-index vs satisfaction). Advice: **follow your heart**.

12. Biotechnology (L43, Prof. Sathyanarayana Gummadi)

- **Areas:** roots in chemical engineering (bioprocess) and basic science. New: therapeutic proteins, monoclonal antibodies, vaccines, **biosimilars** (e.g. erythropoietin), bioinformatics, large genome data.
- **Starting out:** students lack **hands-on accuracy**; first year trained with seniors. Industry is smaller (high investment, trials, regulation); bioprocess graduates place best; start-ups (diagnostic kits, Sea6 Energy). Academia: more publications, post-doc abroad.
- **Progress signs:** sends an interesting paper, proposes new experiments, offers to write a grant. He meets scholars **daily**.

- **Advice:** be a **team player**, identify the problem fast, keep a clear **lab record (record failures)**, **store raw data**, regular hours, read weekly.

13. Chemistry (L44, Prof. B. Viswanathan)

- **Areas:** synthesis lives on, but science must **deliver materials**; main fields **energy, materials, environment**. **MOFs** and mesoporous solids crack heavy oil ("bottom of the barrel"); storage must match petrol (500–600 km); fuel cells need a better **oxygen reduction electrode** (N, S, B doped carbon vs platinum); **DFT** predicts materials before making them.
- **Starting out:** analysis now at atomic level (XPS, UPS) and computing are scarce; students are as capable but more distracted.
- **Industry and jobs:** refineries, drugs in one step, multinationals in India; no worries for 20–30 years.
- **Guide and success:** interactive classes; guide always available, equal treatment (**guru-shishya**). Think unconventionally (**Haber's ammonia**, CO_2 to chemicals). Embrace mathematics as a language.

14. Electrical Engineering (L45, Prof. Deepa Venkitesh)

- **Areas:** power systems and electronics, VLSI, **photonics** (electronics controls electrons, photonics controls photons), communication and signal processing. New: smart grid, solar direct-DC, DC–DC converters, MEMS to NEMS, plasmonics, silicon photonics, smart-city sensors, flexible electronics.
- **Starting out:** exams test only the base of **Bloom's taxonomy** (remember, apply); research needs **create**. Weak maths and problem formulation; little hands-on practice; lack of **ownership** of instruments.
- **Jobs and success:** MS to PhD, PhD to post-doc, industry (Tejas Networks, GE), start-ups. Success = the student brings **ideas the guide had not thought of**.
- **Guide and advice: at least weekly** (more means dependence, less means drift); peer seminars. Work on maths and physics; research is **not time-bound**; teaching helps research.

15. Mathematics (L46, Prof. Arindama Singh)

- **Areas:** geometry, analysis, algebra (old areas never die); fluid dynamics is fading from maths departments. New: numerical analysis, numerical linear algebra, discrete mathematics, theory of computation, image processing.
- **Starting out:** intake is almost always M.Sc. maths; coursework equalises varied B.Sc. standards over 1–2 years.
- **Industry:** Indian industry rarely asks for maths; CS-type graduates go to R&D.
- **Success: solved a long-standing problem**, or **created a new one**, plus papers in good journals; supervisor also checks peripheral knowledge.
- **Guide and advice:** ideally **every day over coffee** (maths is "a linguistic entity"); curious, never satisfied with the surface, but the habit of **reinventing everything** irritates industry. Produce something new beyond the curriculum.

16. Physics (L47, Prof. R. Nirmala)

- **Areas:** optics, atomic and molecular, experimental solid state, non-linear dynamics, theoretical and computational. New: particle physics (after the **Higgs boson**), cosmology, soft matter and biophysics, **quantum computation**, nanoscience (graphene), **quantum phase transitions**, strongly correlated electrons.
- **Starting out:** strengthen core physics; every PhD scholar takes **Foundations in Theoretical Physics** and **Foundations in Experimental Physics**.
- **Industry and jobs:** photovoltaics, organic and oxide electronics, **spintronics**. Most common next step is a **post-doc**, practically mandatory for teaching at NITs, IISERs, IITs, IISc.
- **Success:** can **create a problem and a method**, publishes, talks confidently, collaborates, and can **articulate the problem so it gets funded**.
- **Guide and advice:** she meets scholars **daily**, plus a weekly formal group meeting. Hard work, continuous literature updates, attend colloquia, speak and write well.

17. Discussion with research scholars (L48)

- **Expected vs real:** research needs **patience**; the **goal emerges as you work** (no SOPs, you build the structure); UG labs verify known theory, research builds new theory. **You need interest, patience, hard work, not genius.**
- **Hardest parts:** self-motivation (no deadlines, slow output), isolation after the "pack culture" of UG, **questioning everything**, finding a solvable problem, using freedom well.
- **Conferences:** collaboration and inspiration; work **6–12 months ahead of papers**; broader feedback; **visibility** for post-doc and PhD applications; plan ahead and invite target professors to your talk.
- **After graduation:** academia, industry R&D (MS placement at IIT Madras rose from **50% to 71%** in two years), start-ups, bridging academia and industry.
- **How they changed:** self-responsibility, questioning, working under constraints, humility, patience, observing instead of arguing.
- **Advice:** start early (direct PhD done by ~27); persevere; **accept failure and recognise success** among it; need not know the exact problem when joining an MS; unsure about a PhD, do the master's or industry first; area and guide matter; tinker beyond coursework from second year.

18. Research-design basics the assignment tests ⭐ (outside the lectures)

No talk teaches these; they are standard research-methods facts, so remember them as facts. - **Order of work:** research starts with a **research question**; the **hypothesis** (a tentative answer), design and analysis plan follow. - **Participatory research:** people studied are co-researchers and experts on their situation; knowledge is power; a collective inquiry. Its purpose is **empowerment and action**, not only knowledge. - **Depth of research** is judged by its **objectives** (not title, duration or cost). - **Interdisciplinary research** = integrating knowledge from several disciplines to solve one problem (a theme in every talk). - **Dissertation core:** Introduction; Literature Review; Research Methodology; Results; Discussions and Conclusions. - **Strong objective or test:** valid (measures what it claims), reliable (consistent), impartial or objective (free of bias), and usable (practical). - **Reference sources:** encyclopedia = descriptive information; dictionary = word meanings; directory = names and addresses; bibliography = list of publications. - **Facts:** an electron microscope used in reverse can write tiny patterns (**electron-beam lithography**); Mahabalipuram is a **Pallava** monument.

2. Top traps (most likely first)

1. **Research question comes first**; a hypothesis comes after it.
2. For "which is **false**" items, select the false statement: participatory research is **not** only about producing knowledge.
3. "All of the above" wins when each option is a true criterion (valid, reliable, impartial; reliability, usability, objectivity).
4. **Dissertation order** starts with Introduction and Literature Review; "executive summary" belongs to business reports.
5. **Depth** is judged by objectives, never by title, duration or spending.
6. **Interdisciplinary** aims to integrate disciplines, not oversimplify.
7. Encyclopedia describes; dictionary gives meanings; directory lists addresses.
8. **No single meeting rule:** daily (Mechanical, Biotech, Physics, ideally Maths), about weekly (Civil, Aerospace, Ocean, Electrical minimum, HSS one hour), fortnightly (Metallurgy, CSE at least), fortnightly to monthly at the start (Management).
9. **Success is not only publications**: independence, new ideas, solving or creating a long-standing problem, impact, personal satisfaction.
10. **Analysis vs design:** most departments teach analysis, Engineering Design teaches the design process.
11. About **90% of civil PhDs teach**; GM's B.Tech / Master's / PhD sentences; project work vs research.
12. Lecturer slips: "phrenology" is **phonology**; "heart tearing" is **hot tearing**.

3. Assignment questions with solutions

Q1. In order to pursue the research, which of the following is priorly required? *(Q1)* - A. Developing a research design B. Formulating a research question C. Deciding about the data analysis procedure D. Formulating a research hypothesis

Solution (card 18, outside the lectures): the question says what you want to find out, and the hypothesis, design and analysis plan are built to answer it. **Answer: B.**

Q2. Which one among the following statements is false in the context of participatory research? *(Q2)* - A. It recognizes knowledge as power B. It is a collective process of inquiry C. It emphasizes people as experts D. Its sole purpose is the production of knowledge

Solution (card 18): A, B and C describe it correctly. It also aims at empowerment and change, so "sole purpose is knowledge" is false. **Answer: D.**

Q3. How to judge the depth of any research? *(Q3)* - A. By research title B. By research duration C. By research objectives D. By total expenditure on research

Solution (card 18): objectives show how deep the questions go; title, time and cost are surface features. **Answer: C.**

Q4. What is the main aim of interdisciplinary research? *(Q4)* - A. To over simplify the problem of research B. Integrating knowledge from multiple disciplines to solve a problem C. To create a new trend in research methodology D. To reduce the emphasis on a single subject in the research domain

Solution (cards 1–17 show it in every department): complex problems need several fields combined. **Answer: B.**

Q5. Electron microscopes can be used in reverse to write in extremely small areas. *(Q5)* - A. True B. False

Solution (card 18, a fact from general science): a focused electron beam can draw patterns a few nanometres wide (electron-beam lithography). **Answer: A (True).**

Q6. What are the core elements of a dissertation? *(Q6)* - A. Introduction; Data Collection; Data Analysis; Conclusions and Recommendations B. Executive Summary; Literature Review; Data Gathered; Conclusions; Bibliography C. Research Plan; Research Data; Analysis; References D. Introduction; Literature Review; Research Methodology; Results; Discussions and Conclusions

Solution (card 18): only D has the standard order and includes the literature review and methodology. **Answer: D.**

Q7. How can we enhance the research objective? *(Q7)* - A. By making it more valid B. By making it more reliable C. By making it more impartial D. All of the above

Solution (card 18): each of A, B and C strengthens an objective. **Answer: D.**

Q8. A common test in research demands much priority on *(Q8)* - A. Reliability B. Useability C. Objectivity D. All of the above

Solution (card 18): a good test is consistent, practical and gives the same result whoever marks it. **Answer: D.**

Q9. An appropriate source to find out descriptive information is *(Q9)* - A. Bibliography B. Directory C. Encyclopedia D. Dictionary

Solution (card 18): descriptive, explanatory information comes from an encyclopedia. **Answer: C.**

Q10. Mahabalipuram in south-east India, pre-eminently testimony to the …….. civilization *(Q10)* - A. Pallavas B. Mauryas C. Guptas D. Cholas

Solution (general knowledge, not taught): its rock-cut temples and Shore Temple were built by the Pallava dynasty in the 7th and 8th centuries; Cholas are linked with Thanjavur. **Answer: A.**